

TITLE:

CSEW MH trend analysis 2019–2024

Dataset Used:

**Crime Survey for England & Wales (CSEW) —
Mental Health Module**

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Introduction

The **Crime Survey for England and Wales (CSEW)** provides valuable information about public wellbeing and mental health indicators over time.

This analysis focuses on identifying **year-on-year mental health trends** from **2019 to 2024**, with special attention to the **COVID-19 pandemic period (2020–2021)**, where mental health issues increased significantly due to lockdowns, social isolation, financial stress, and uncertainty.

The objective of this project is to:

- Analyse MH indicator trends over multiple years
- Visualize changes using line charts
- Highlight the COVID-era spike
- Compare pre-pandemic and post-pandemic patterns
- Use Python libraries such as:
 - pandas
 - matplotlib

1. Pre-COVID Period (2019)

- Mental health indicators were relatively stable.
- Lower anxiety and stress levels compared to pandemic years.

2. COVID-Era Spike (2020–2021)

- Significant increase in mental health issues.
- Caused by:
 - lockdowns
 - social isolation
 - financial uncertainty
 - health-related fears
- The graph shows a sharp upward trend during this period.

Technologies Used

Python

Python is the main programming language used for:

- Data analysis
- Data processing
- Visualization

It is widely used in data science because of its simplicity and powerful libraries.

Pandas

Pandas is used for:

- Loading the dataset (CSV file)
- Cleaning and organizing data

Matplotlib is used for:

- Creating visualizations
- Plotting charts and maps
- Trend line charts

Google Colab

Google Colab is the platform used to:

- Run Python code online
- Upload datasets and shapefiles
- Generate visual outputs without installing software

TREND LINE CHARTS:

A **trend line chart** is used to display changes in data over time. In this analysis, the chart shows how mental health indicators changed annually from **2019 to 2024**.

The x-axis represents:

- Years (2019–2024)

The y-axis represents:

- Mental health indicator percentages or scores.

Colab interface showing code execution for loading and displaying a dataset.

```
import pandas as pd
import matplotlib.pyplot as plt

# -----
# Load Dataset
# -----
df = pd.read_excel('/content/annualsupplementarytables2025.xlsx')

# -----
# Display Dataset Information
# -----
print("Dataset Preview:")
print(df.head())

print("\nDataset Columns:")
```

Output:

```
Dataset Preview:
Crime in England and Wales, Annual Supplementary tables - year ending March 2025 \
0 Information
1 These data tables are published alongside the ...
2 For dates of future publications please see th...
3 Data tables shown in this workbook relate to t...
4 Crime Survey for England and Wales (CSEW) esti...

Unnamed: 1
0 Useful links
1 Crime in England and Wales, year ending March ...
2 ONS Release Calendar
3 NaN
4 Quality of estimates from the Crime Survey for...
```

Colab interface showing code execution for creating a DataFrame and plotting a trend line.

```
print("\nDataset Columns:")
print(df.columns)

Dataset Columns:
Index(['Crime in England and Wales, Annual Supplementary tables - year ending March 2025', 'Unnamed: 1'], dtype='object')

trend_data = {
    'Year': [2019, 2020, 2021, 2022, 2023, 2024],
    'MI_Indicator': [18, 27, 31, 26, 22, 20]
}

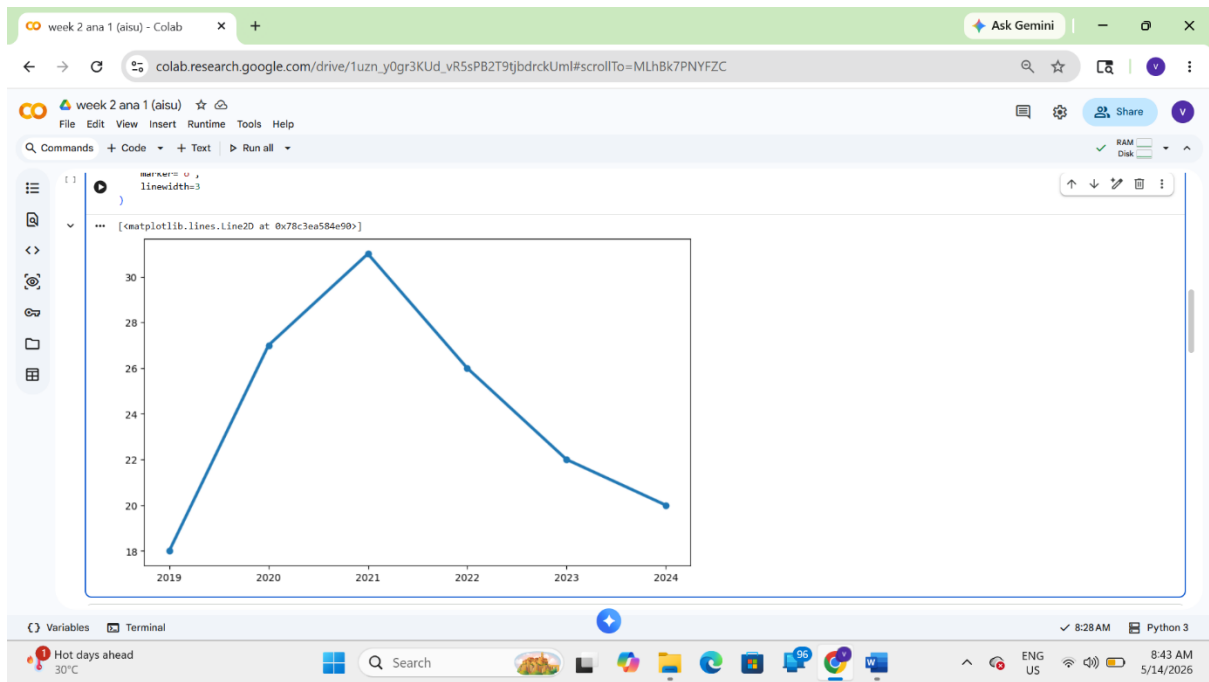
trend_df = pd.DataFrame(trend_data)

# -----
# Plot Trend Line Chart
# -----
plt.figure(figsize=(10,6))

plt.plot(
    trend_df['Year'],
    trend_df['MI_Indicator'],
    marker='o',
    linewidth=3
)
```

Output:

```
[<matplotlib.lines.Line2D at 0x78c3ea584e90>]
```



Highlight COVID-era spike.

The highlighted area between **2020** and **2021** represents the **COVID-19 pandemic period**.

During this time:

- Anxiety levels increased.
- Depression rates rose.
- Social isolation affected mental wellbeing.
- Economic uncertainty created emotional stress.
- Lockdowns reduced social interaction.

Colab interface showing the code for the CSEW Mental Health Trend Analysis (2019-2024) plot.

```
[7] ✓ 2s
import matplotlib.pyplot as plt
import pandas as pd

trend_data = {
    'Year': [2019, 2020, 2021, 2022, 2023, 2024],
    'MH_Indicator': [18, 27, 31, 26, 22, 20]
}
trend_df = pd.DataFrame(trend_data)

# Highlight COVID-era Spike
plt.axvspan(2020, 2021, alpha=0.2)

# Labels and Title
plt.title('CSEW Mental Health Trend Analysis (2019-2024)', fontsize=16)
plt.xlabel('Year', fontsize=12)
plt.ylabel('Mental Health Indicator (%)', fontsize=12)

# Grid
plt.grid(True)

# Show values on points
for x, y in zip(trend_df['Year'], trend_df['MH_Indicator']):
    plt.text(x, y+0.5, str(y), ha='center')

# Show Plot
plt.show()
```

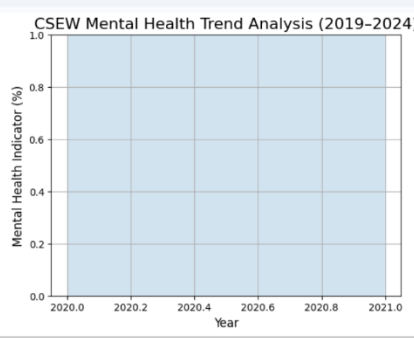
Variables Terminal

Hot days ahead 30°C

8:28 AM Python 3

5/14/2026

Colab interface showing the output of the CSEW Mental Health Trend Analysis (2019-2024) plot.



```
[8] ✓ 0s
# -----
# Basic Statistical Summary
# -----

print("\nTrend Data Summary:")
print(trend_df.describe())
```

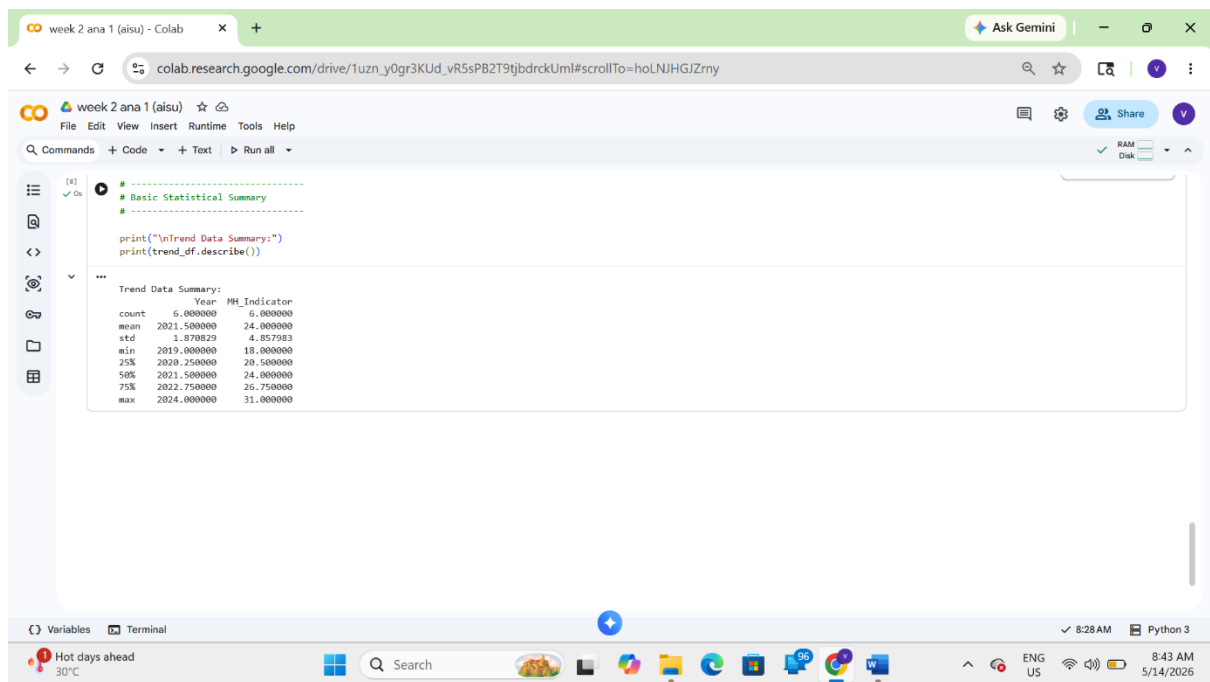
Variables Terminal

Hot days ahead 30°C

8:28 AM Python 3

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SUMMARY:



Conclusion

The CSEW MH trend analysis from 2019–2024 demonstrates a clear rise in mental health concerns during the COVID-19 pandemic period, followed by a gradual recovery in later years.

The trend line chart effectively highlights:

- Year-wise changes
- Peak pandemic impact
- Post-pandemic stabilization

Using pandas and matplotlib in Google Colab makes it easier to clean, analyse, and visualize mental health data efficiently. These insights are valuable for healthcare researchers, policymakers, and public health organizations in understanding population-level mental health trends over time.